

Introduction to Contract Theory and Auctions

Reserve Prices

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

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Reserve Price

A **reserve price** $r > 0$, is a price that the auctioneer puts such that the object is not sold if the winning bid in the auction is less than r . Thus, the seller *reserves the right* to not sell the object if the selling price determined by the auction is less than the threshold amount r . The purpose of such a reserve price is to **discriminate or exclude potential buyers with low valuations**.

Reserve Prices in the SPA

- Since the price needs to be at least greater or equal to r , no bidder with value $v < r$ can make a positive profit in the auction.
- Only bidders with values $v \geq r$ will participate.
- In terms of bidding behavior, it is still a **weakly dominant strategy** to bid your own value, even with a reserve price.
- Consider a bidder with value $v = r$, then the expected payment for this bidder is

$$m(r) = G_1^{(n-1)}(r)r$$

$\hookrightarrow \Pr(Y_1^{(n-1)} \leq r) = \Pr(\text{Winning when } v=r)$

- For a bidder with a value $v > r$ the expected payment is now

$$m(v) = G_1^{(n-1)}(r)r + \int_r^v y g_1^{(n-1)}(y) dy$$

$\downarrow$
 $Y_1^{(n-1)} < r < v$ $r < Y_1^{(n-1)} < v$

Reserve Prices in the FPA

- No bidder with value $v < r$ can make a positive profit.
- Consider $\beta : \mathcal{V} \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$, a symmetric equilibrium of the FPA, then we must have that $\beta(r, r) = r$. Why? $\hookrightarrow \beta(\text{value, reserve price}) \mapsto \text{bid}$
- If $v > r$, we have that

$$\beta(v, r) = E[\max\{Y_1^{(n-1)}, r\} | Y_1^{(n-1)} \leq v]$$

$$= \frac{1}{G_1^{(n-1)}(v)} \int_0^v \max\{y, r\} g_1^{(n-1)}(y) dy$$

$$= \frac{1}{G_1^{(n-1)}(v)} \left[\int_0^r r g_1^{(n-1)}(y) dy + \int_r^v y g_1^{(n-1)}(y) dy \right] = \frac{1}{G_1^{(n-1)}(v)} \left[r G_1^{(n-1)}(r) + \int_r^v y g_1^{(n-1)}(y) dy \right]$$

$$= \frac{r G_1^{(n-1)}(r)}{G_1^{(n-1)}(v)} + \frac{1}{G_1^{(n-1)}(v)} \int_r^v y g_1^{(n-1)}(y) dy = \beta(v, r)$$

- Thus, when $v \geq r$: $m(v, r) = \cancel{G_1^{(n-1)}(v)} G_1^{(n-1)}(v) \cdot \beta(v, r)$

$$\begin{aligned}
 m_i^I(u, r) &= G_i^{(n-1)}(u) \beta(u, r) \\
 &= r G_i^{(n-1)}(r) + \int_r^u y g_i^{(n-1)}(y) dy \quad \text{if } u \geq r
 \end{aligned}$$

L, Which is the same as in the SPA.

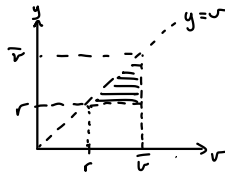
$$m_i^I(r, r) = r G_i^{(n-1)}(r) + \int_r^r y g_i^{(n-1)}(y) dy = r G_i^{(n-1)}(r)$$

$$\begin{aligned}
 m_i^I(u, r) &= \begin{cases} r G_i^{(n-1)}(r) + \int_r^u y g_i^{(n-1)}(y) dy & \text{if } u \geq r \\ 0 & \text{if } u < r \end{cases} \\
 &= \left(r G_i^{(n-1)}(r) + \int_r^u y g_i^{(n-1)}(y) dy \right) \times I_{\{u \geq r\}}(u)
 \end{aligned}$$

Revenue Effects of Reserve Prices

Notice that, irrespective of the auction format, the interim expected payment with a reserve price r is:

$$m(v, r) = \left(rG_1^{(n-1)}(r) + \int_r^v yg_1^{(n-1)}(y)dy \right) I_{v \geq r}(v).$$



Thus,

$$\begin{aligned} \int_0^{\bar{v}} m(v, r) f(v) dv &= E[m(V, r)] = \int_0^{\bar{v}} \left(rG_1^{(n-1)}(r) + \int_r^v yg_1^{(n-1)}(y)dy \right) I_{v \geq r}(v) f(v) dv \\ &= \int_r^{\bar{v}} \left(rG_1^{(n-1)}(r) + \int_r^v yg_1^{(n-1)}(y)dy \right) f(v) dv \\ &= \int_r^{\bar{v}} r G_1^{(n-1)}(r) f(v) dv + \int_r^{\bar{v}} \left(\int_r^v yg_1^{(n-1)}(y) dy \right) f(v) dv \\ &= r G_1^{(n-1)}(r) (1 - F(r)) + \int_r^{\bar{v}} yg_1^{(n-1)}(y) \left(\int_y^{\bar{v}} f(v) dv \right) dy \\ &= r G_1^{(n-1)}(r) (1 - F(r)) + \int_r^{\bar{v}} yg_1^{(n-1)}(y) (1 - F(y)) dy \end{aligned}$$

Optimal Reserve Prices

- Suppose that the seller attaches a value $v_0 \in [0, \bar{v})$ to the object.
- The seller will never use $r < v_0$.
- The expected revenue of a reserve price $r \geq v_0$ is

$$\begin{aligned}\Pi_0(r) &= n E[m(V, r)] + G_1^{(n)}(r)v_0 = n E[m(V, r)] + F(r)^n v_0 \\ &= n \left\{ r G_1^{(n-1)}(r) (1 - F(r)) + \int_r^{\bar{v}} y g_1^{(n-1)}(y) (1 - F(y)) dy \right\} + F(r)^n v_0\end{aligned}$$

- In order to the optimal reserve price r^* , we need to find $\frac{d\Pi_0}{dr}$

$$\begin{aligned}\Pi_0 &= n r G_1^{(n-1)}(r) - n r G_1^{(n-1)}(r) F(r) + n \int_r^{\bar{v}} y g_1^{(n-1)}(y) (1 - F(y)) dy + F(r)^n v_0 \\ \frac{d\Pi_0}{dr} &= n (G_1^{(n-1)}(r) + r g_1^{(n-1)}(r)) - n (r G_1^{(n-1)}(r) f(r) + [G_1^{(n-1)}(r) + r g_1^{(n-1)}(r)] F(r)) \\ &\quad - n r g_1^{(n-1)}(r) (1 - F(r)) + n F(r)^{n-1} f(r) v_0 \\ &= n G_1^{(n-1)}(r) + \cancel{n r g_1^{(n-1)}(r)} - n r G_1^{(n-1)}(r) f(r) - n G_1^{(n-1)}(r) F(r) - \cancel{n r g_1^{(n-1)}(r) F(r)} \\ &\quad - n r \cancel{g_1^{(n-1)}(r)} + \cancel{n r g_1^{(n-1)}(r) F(r)} + n F(r)^{n-1} f(r) v_0\end{aligned}$$

$$\begin{aligned}
 \frac{d\pi_0}{dr} &= n G_1^{(n-1)}(r) - n G_1^{(n-1)}(r) F(r) - n r G_1^{(n-1)}(r) f(r) \\
 &\quad + n G_1^{(n-1)}(r) f(r) v_0 \\
 &= n G_1^{(n-1)}(r) [1 - F(r) - r f(r) + f(r) v_0] \cdot \frac{1 - F(r)}{1 - F(r)}
 \end{aligned}$$

Define the Hazard Rate $\lambda(x) = \frac{f(x)}{1 - F(x)}$

$$\begin{aligned}
 &= n G_1^{(n-1)}(r) (1 - F(r)) [1 - r \lambda(r) + \lambda(r) v_0] \\
 &= n G_1^{(n-1)}(r) (1 - F(r)) [1 - (r - v_0) \lambda(r)]
 \end{aligned}$$

$$\text{F.O.C: } n G_1^{(n-1)}(r^*) (1 - F(r^*)) [1 - (r^* - v_0) \lambda(r^*)] = 0$$

$$\Rightarrow (r^* - v_0) \lambda(r^*) = 1 \Leftrightarrow r^* - \frac{1}{\lambda(r^*)} = v_0 \Leftrightarrow r^* - \frac{1 - F(r^*)}{f(r^*)} = v_0$$

$$r^* - \frac{1 - F(r^*)}{f(r^*)} = v_0$$

Suppose $v_0 = 0 \Rightarrow r^* = \frac{1 - F(r^*)}{f(r^*)}$

$\hookrightarrow V_i \sim U[0,1], n=2, v_0=0$

$$r^* = 1 - r^* \Rightarrow r^* = 1/2$$

Without Reserve price: Revenues = $1/3$

With $r^* = 1/2$: Revenues $\frac{5}{12}$ (Check as HW )